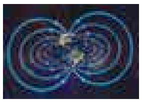
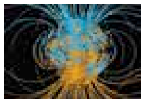


The north is moving
The movement of the material within the planetary core is causing the magnetic north to shift positions. **https://dgit.in/jun20-70**



Geomagnetic reversal
Why you probably should not be worried about the poles of the Earth getting interchanged. **https://dgit.in/jun20-71**

century. Song Dynasty scholar, Shen Kuo, wrote 'Dream Pool Essays' in 1086 AD which described how a needle was magnetised by rubbing its tip to lodestone. The magnetised needle was attached to a single strand of silk with a bit of wax. The scholar documented that the needle, when suspended in a windless area, would always point to the south. The Song Dynasty military used them for land navigation in 1040-1044 AD and they were utilised for sea navigation during 1111-1117 AD. The Chinese also crafted wet magnetic compasses which had a magnetised needle floating in a bowl of water.

The magnetic compass made its way over to Europe in 1190 AD. Alexander Neckam, an English scholar, documented the use of a magnetised needle to find directions at sea in De naturis

Model of the Chinese 'south pointing' compass (206BC-220AD)

rerum (On the Natures of Things), which was written in 1190. By the 1300s, Europe and the Middle East had adopted the magnetic compasses. However, there is still a debate over whether the Europeans independently created the magnetic compasses many centuries after the Chinese, or if the Chinese introduced it to the people in the Middle East who then shared their knowledge with the Europeans. During this period, trade between the East and the West began booming. Sailing season, which was previously restricted to the months from June to September, now started off as early as late January and ended in early December! Commercial travel and trade increased in the Middle East, Europe and also China which accounted for an economic boost.

However, travellers began encountering a challenge with these compasses. The problem of magnetic variation (magnetic declination) was faced, where there is a clear angular difference between the geographic or 'true' north and magnetic north (where the magnetic compass needle points). The difference was relatively small across the Mediterranean, however, in the Atlantic, especially the northern latitudes, the difference was pretty sizable. The first documentation of magnetic declination was in Kuan Shih Ti Li Chih Meng (Mr Kuan's Geomantic Instructor), which dates back to 880 AD.

In 1698, Sir Edmund Halley, with the support of England's Royal Society and the Admiralty, set out on long expeditions to study and measure Earth's magnetic variations across the northern and southern regions of the Atlantic Ocean. The data was beneficial to the English Navy and in 1701, Halley revealed the world's first isogonic chart which details how the angle between magnetic north and true north varies in different areas across the Atlantic Ocean.

MORE DISCOVERIES

The dry mariner's compass features a freely pivoting needle on a pin enclosed within a tiny box with a glass cover and a wind rose (also known as the compass card). Such pivoting needles enclosed in glass boxes were already described by a couple of scholars in the past, however, Flavio Gioja, an Italian marine pilot, is credited with perfecting this compass in the 1300s. He came up with the idea to suspend the needle over a compass card to give it a familiar appearance. During this time, both dry and floating compasses were used actively in Europe. When compasses arrived in the Muslim countries in the 1300s, they were also used for astronomical purposes, in addition to naval navigation. Magnetic compasses were used as the Qibla indicator to find the direction to Mecca for daily prayers. Syrian astronomer, Ibn al-Shatir also improved upon this indicator by combining a universal sundial and a magnetic compass.

After the invention of the dry compasses, another variant of the magnetic compass saw the light of day – the bearing compass. It allows the taking of bearings of objects by aligning the lubber line of the compass to the object. Another invention was the surveyor's compass which can accurately measure the heading of landwards and horizontal angles which helped in mapmaking. These compasses became commonplace by the early 18th century and are even displayed in the 1728 Cyclopaedia. As time passed, the bearing compass became smaller and smaller until it could be held in the hand. A patent was granted for a hand compass in 1855 which featured a viewing prism and a lens that allowed users to see the heading of geographical landmarks. Later, the liquid compass was created in which the magnetised needle is dampened to avoid excessive wobble or swing, which improves readability. Sir Edmund Halley introduced a rudimentary design of this in 1690. However, early liquid compasses were heavy and bulky. The first practically-designed liquid compass was patented by Francis Crow in 1813. These compasses were used occasionally by the Royal Navy from the 1830s to 1860. In 1861, American physicist and inventor, Edward Samuel Ritchie patented a vastly improved liquid marine compass which was extensively used by the US Navy as well as the Royal Navy, later in 1908.

Modern-day compasses are based on the same principles as those from over 2000 years ago. Modern navigational compasses feature a magnetised needle encased in a fluid-filled capsule. The fluid acts as a deterrent to stop the needle from oscillating back and forth. Many modern day compasses incorporate a baseplate and protractor tool. Some even have map and romer scales to measure distances on maps, sighting mechanisms such as mirrors and prisms to take bearings of distant objects, and more. d